

TALAT KARASAKAL

Computer Engineering Student

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ABOUT ME

I find explaining why a system was built the way it was harder — and more valuable — than getting it to run. I study computer engineering and work mostly on desktop and web applications and image processing. The part of a task that interests me most is where the options get compared: which model, which library, which data setup was chosen and why. My goal is to contribute, through software and systems development, to Türkiye reaching global standards in artificial intelligence.

EDUCATION

Izmir University of Economics — BSc in Computer Engineering

Izmir, Türkiye | 09/2022 – 06/2027 (expected) | English-medium instruction | 50% merit scholarship

PROJECTS

Web-Based Survey Application

Probel Yazılım internship, solo | Python, Flask, SQLAlchemy, SQLite | 08/2026 – 09/2026

- Prepared a 49-page requirements and design document following IEEE 830 and IEEE 1016: 66 functional requirements (77 including sub-items), 20 use cases and a traceability matrix.
- Selected Flask, SQLAlchemy and SQLite, ruling out Django with a documented rationale; defined a nine-entity data model and a co-ownership model with five separate permissions.
- Verified that client-side validation could be bypassed and decided to enforce all checks server-side; identified a privilege-escalation chain via admin permissions and restricted admin grants to the primary owner.

Nexa — Smart Road Safety with 5G and AI (TEKNOFEST 2026, Turkcell)

3-person team, AI owner | Python, YOLOv8, OpenCV, EasyOCR, Docker | 03/2026 – 07/2026

- Made and validated the technical decisions for an inference pipeline detecting vehicle type, licence-plate region, colour, driver violations and seat occupancy from a fixed roadside camera using a single 21-class YOLOv8s model.
- Remapped 10 open-source datasets onto the competition's 21 official classes and annotated team-recorded footage for uncovered classes, assembling a 32,425-image dataset while preventing leakage from consecutive frames at the split stage.
- Benchmarked YOLOv8s against YOLOv8m on two GPUs and selected YOLOv8s: 0.701 overall mAP@0.5, 0.896 for plate-region detection, ~12 ms per frame. The team cleared the Preliminary Design Report stage with 85.67/100.

Exam Scheduling and Management System

5-person course project (SE 302) | Java, JavaFX, SQLite, Gradle | 10/2025 – 12/2025

- Owned the interface and input-handling responsibility in a desktop application automating exam-week scheduling, and designed the entire interface. Recommended Java to the team, which accepted it.
- Proposed parallelising the scheduling computation, which substantially reduced runtime. Proposed a persistence approach storing recently opened files and filters, removing the need to recompute on every launch.

Integrated Assignment Environment

5-person course project (CE 316) | Java, JavaFX, SQLite | 05/2026 – 06/2026

- Selected the execution-engine module of a five-module team for an application that compiles and runs student submissions in isolation and compares output with expected results; the module covered process management, timeouts, runtime measurement, compiler path checks and error classification.
- A timeout defect where only the parent process was terminated while child processes kept running was fixed. Three unit-test files were added for the module; one caught a defect where submissions failing at runtime were counted as passing.

Discrete Event Simulation

4-person course project (SE 116) | Java | 04/2024 – 05/2024

- Worked on the input-parsing layer of a Java application that parses and validates two configuration files and reports station queues and delay metrics; identified and fixed parsing-loop defects in the JobFileParser and WorkflowFileParser classes.

EXPERIENCE

Probel Yazılım — Computer Engineering Intern

Izmir, Türkiye | 08/2026 - 09/2026 | IT and R&D Department

- Completed the mandatory summer internship at a healthcare information systems company, running a web-based survey application end to end as a standalone R&D project independent of the company codebase.
- Studied SDLC and architecture topics assigned by the mentor: modular monolith, dependency inversion, CQRS, hexagonal architecture (ports and adapters) and SOLID.

Izmir University of Economics — Student Assistant, Corporate Communications Directorate

Izmir, Türkiye | 08/2025 - 12/2025, 02/2026 - 07/2026, 10/2026 - Present

- Supported on-site coordination of large-scale events in the Events Unit, including the academic year opening, commencement ceremonies, the Long Service Awards and the 25th anniversary celebrations; represented the university at national commemorations.

SCA Social — Project Management Intern (Volunteer)

Istanbul, Türkiye (Remote) | 01/2026 - 02/2026

- Prepared annual budget planning and cost tables for a defence-industry context; designed an AI-based simulation approach for early identification of risks in construction projects.
- Turned data protection (KDPL) processes into formal documentation through case analysis; produced a Project Initiation Document and Gantt chart covering a project's scope, resource and time planning.

Izmir University of Economics — Admissions and Outreach Assistant

Izmir, Türkiye | 07/2024 - 09/2024, 07/2025 - 08/2025

- Advised prospective students and parents in the call-centre and transfer units, and ran campus tours.

VOLUNTEER EXPERIENCE

IEU Software Community — Member, Audit Board Member

Membership 10/2024 - Present | Audit Board 09/2025 - 08/2026

Industrial Systems Society — Member

10/2024 - Present

CERTIFICATIONS AND TRAINING

AI and Data: Introduction to Modern AI, Introduction to Data Science (Cisco, 2024) · AI Fluency for Students, Claude Platform 101, Introduction to Model Context Protocol, Model Context Protocol: Advanced Topics (Anthropic, 2026)

Project Management: Agile Project Management, Project Management with MS Project (BTK Academy, 2026)

Quality and Information Security: ISO 9001:2015 Quality Management System (Sigmacert Global, 2026) · Information Security Fundamentals (IUE, 2024)

Industry and Tools: Career and Competency Meetings (Defence Industry Academy, 2024–2026) · Kuika Designer and Builder Course (Kuika Academy, 2025) · App Designer Workshop (Figes, 2025)

LANGUAGES

Turkish — Native · English — B2 · German — A2

TECHNICAL SKILLS

Programming Languages: Java, Python

Frameworks and Libraries: JavaFX, Flask, SQLAlchemy

Databases: SQLite

Tools: Git / GitHub, Gradle

Process: Requirements engineering (IEEE 830 / IEEE 1016), project management and planning, event organisation and coordination